Evaluating Limits

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Direct substitution and algebraic techniques**

Evaluate each limit:

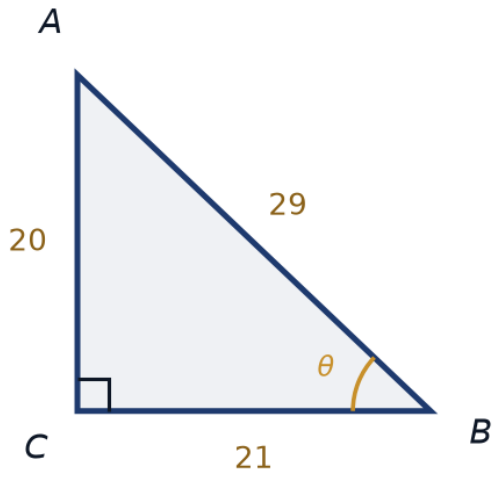
a $\lim_{x \to 1} (3x^2 - 2x + 5)$

b $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$

c $\lim_{x \to 0} \frac{\sqrt{x + 4} - 2}{x}$

2 **[Foundational]** • **Target: Trigonometric and special limits**

Using $\lim_{x \to 0} \frac{\sin x}{x} = 1$, evaluate:



a $\lim_{x \to 0} \frac{\sin 5x}{x}$

b $\lim_{x \to 0} \frac{1 - \cos x}{x}$

c $\lim_{x \to 0} \frac{\sin 3x}{\sin 5x}$

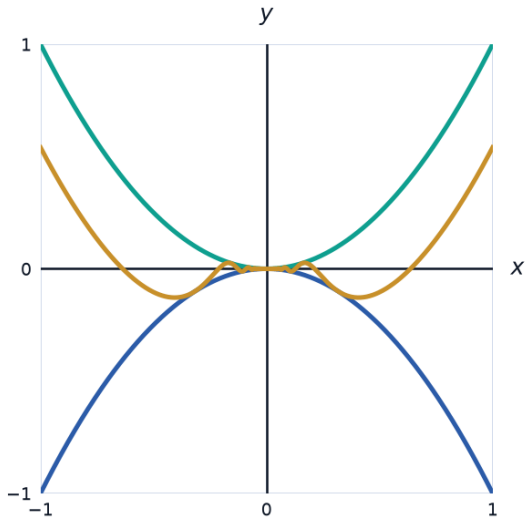
3 **[Foundational]** • **Target: Limits that fail to exist and the squeeze theorem**

Explain why $\lim_{x \to 0} \sin\left(\frac{1}{x}\right)$ does not exist.

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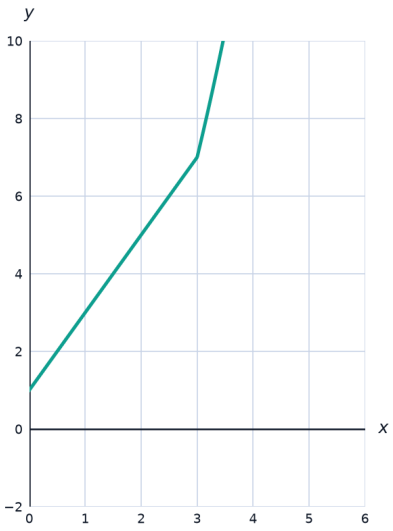
4 [Foundational] • Target: Limits that fail to exist and the squeeze theorem

Use the squeeze theorem to evaluate $\lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right)$.



5 [Foundational] • Target: Limits with absolute value and piecewise functions

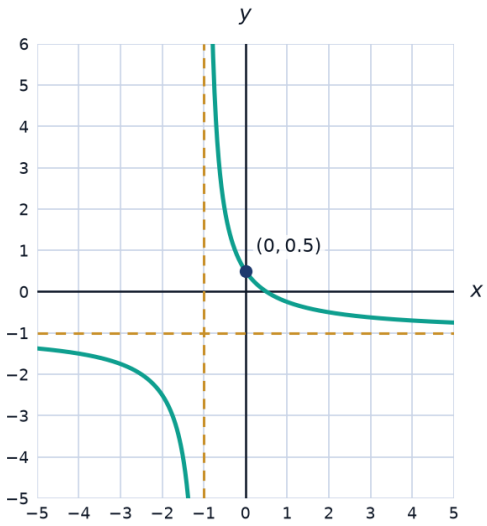
For $f(x) = 2x + 1$ if $x < 3$, and $f(x) = x^2 - 2$ if $x \geq 3$, evaluate $\lim_{x \to 3^-} f(x)$ and $\lim_{x \to 3^+} f(x)$, and state whether $\lim_{x \to 3} f(x)$ exists.



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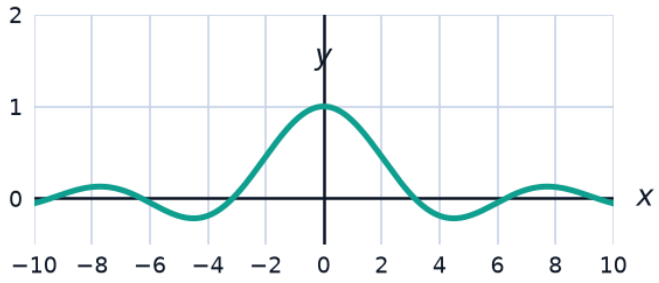
6 [Foundational] • Target: More algebraic limit techniques

Evaluate $\lim_{x \to 2} \frac{x^3 - 8}{x - 2}$ by factoring with the difference-of-cubes formula.



7 [Foundational] • Target: Visualizing a limit numerically and graphically

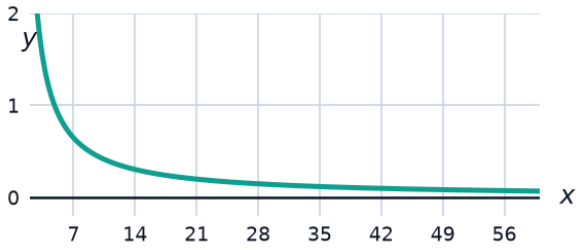
The function $f(x) = \frac{\sin x}{x}$ (undefined at $x = 0$) is graphed below. Use the graph to estimate $\lim_{x \to 0} f(x)$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Limits at infinity with different degree relationships

Evaluate each limit at infinity, based on the relative degrees of numerator and denominator:



a $\lim_{x \to \infty} \frac{4x + 3}{x^2 - 1}$ (lower degree numerator)

b $\lim_{x \to \infty} \frac{x^3 - 1}{2x^2 + 5}$ (higher degree numerator)

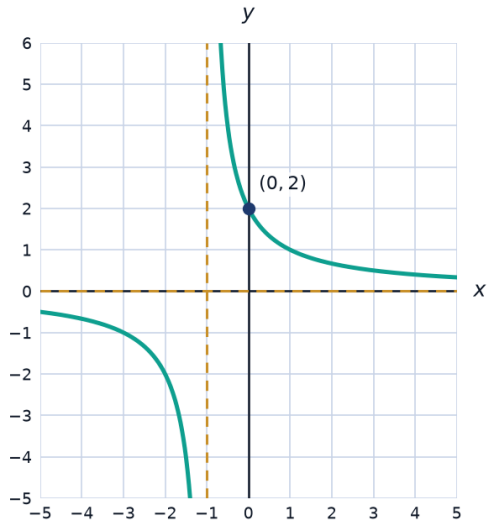
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9 [Intermediate] • Target: Limits requiring conjugate multiplication

Evaluate $\lim_{x\to 9} \frac{x-9}{\sqrt{x}-3}$ by rationalizing the denominator.

10 [Intermediate] • Target: Limits requiring conjugate multiplication

Evaluate $\lim_{x\to\infty} \left(\sqrt{x^2+3x}-x\right)$ by rationalizing.



11 [Intermediate] • Target: Piecewise limits requiring a parameter

For $f(x) = kx + 1$ if $x \le 2$, and $f(x) = x^2 - 3$ if $x > 2$, find k so that $\lim_{x\to 2} f(x)$ exists.

12 [Intermediate] • Target: Limits involving factorable higher-degree polynomials

Evaluate $\lim_{x\to 1} \frac{x^4-1}{x-1}$ by factoring the numerator as a difference of squares twice.

13 [Intermediate] • Target: Limits involving factorable higher-degree polynomials

Evaluate $\lim_{x\to -2} \frac{x^3+8}{x+2}$ using the sum-of-cubes factoring pattern.

14 [Intermediate] • Target: Limits with absolute value and piecewise functions

Level 2 · MediumEvaluate the trigonometric limit: $\lim_{x\to 0} \frac{1-\cos(3x)}{x^2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

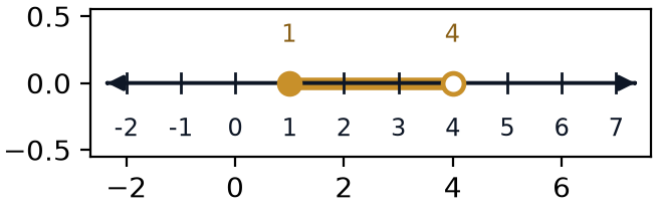
15 [Challenging] • Target: More algebraic limit techniques • Cross-Lesson: 1.01 Precalculus Review for Calculus

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x\to 0} \frac{\sqrt{1+\tan x}-\sqrt{1+\sin x}}{x^3}$.

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16 [Challenging] • Target: Visualizing a limit numerically and graphically • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Level 1 · FoundationsEvaluate the one-sided limit $\lim_{x \to 2^-} \frac{|x-2|}{x^2-4}$.



17 [Challenging] • Target: Limits at infinity with different degree relationships • Cross-Lesson: 11.01 Sequences and Series

Level 3 · AdvancedWater flows into an initially empty tank at a rate $R(t) = 120 - 4t$ liters/hour and leaks out at a rate $L(t) = 3t$ liters/hour for $0 \le t \le 24$ hours. Find the maximum volume of water in the tank and the time at which the tank becomes completely empty.

18 [Challenging] • Target: Limits requiring conjugate multiplication • Cross-Lesson: 12.01 Convergence Tests

Level 3 · Advanced Evaluate the improper integral or establish its divergence: $\int_0^{\infty} xe^{-x/2}\,dx$

19 [Challenging] • Target: Piecewise limits requiring a parameter • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 2 · MediumFind the minimum number of subintervals n required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than 10^{-4}.

20 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int \frac{5x-3}{x^2-2x-3}\,dx$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Direct substitution and algebraic techniques

Consider the function $f(x)$ whose graph is given below. The function is defined piecewise. For $x \le 2$, $f(x) = ax + b$. For $x > 2$, $f(x) = cx + d$. The graph of $f(x)$ passes through the point $(0, 3)$. The limit of $f(x)$ as x approaches 2 exists. Find the value of a.

a. -1

b. 1

c. 2

d. -2

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22 [Foundational] • Target: Trigonometric and special limits

Direct comparison test: if $0 \le f(x) \le g(x)$ for $x \ge a$ and $\int_a^\infty g(x)\,dx$ converges, then $\int_a^\infty f(x)\,dx$:

a. also converges

b. must diverge

c. converges only if $f = g$

d. cannot be determined

23 [Foundational] • Target: Limits that fail to exist and the squeeze theorem

$\int \frac{dx}{\sqrt{4-x^2}}$ evaluated at $x = 2\sin\theta$: after substitution:

a. $\arcsin(x/2) + C$

b. $2\cos\theta d\theta + C$

c. $\sin\theta + C$

d. $d\theta + C$

24 [Foundational] • Target: Limits with absolute value and piecewise functions

$\int_0^1 \frac{1-x^n}{1-x}\,dx = \sum_{k=1}^{n} \frac{1}{k} = H_n$. Evaluating by writing $\frac{1-x^n}{1-x} =$

a. $(1-x)^{n-1}$

b. $n/(1-x)$

c. $\ln(1-x)$

d. $1 + x + \cdots + x^{n-1}$

25 [Foundational] • Target: More algebraic limit techniques

Continuity of piecewise: $f(x) = \{kx, x < 2; x^2 - 2, x \ge 2$ is continuous iff $k =$

a. 1

b. 2

c. 0

d. 3

26 [Foundational] • Target: Visualizing a limit numerically and graphically

Intermediate Value Theorem: If f is continuous on $[1, 4]$, $f(1) = -2$, $f(4) = 5$, must $f(c) = 0$ for some $c \in (1, 4)$?

a. No, not guaranteed

b. Yes, by IVT

c. Only if f is differentiable

d. Only at $c = 2.5$

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27 [Foundational] • Target: Limits at infinity with different degree relationships

Evaluate the one-sided limit $\lim_{x\to 3^-} \frac{|x-3|}{x-3}$.

a. 0

b. ∞

c. -1

d. 1

28 [Foundational] • Target: Limits requiring conjugate multiplication

For $\int \frac{x^2+1}{x(x-1)(x+1)}\,dx$ the degree of numerator vs denominator:

a. improper; use long division

b. proper; partial fractions directly

c. degree equal; long division

d. none

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Piecewise limits requiring a parameter

Trig substitution for $\sqrt{a^2-x^2}$: use $x=$

a. $a\sin\theta$

b. $a\tan\theta$

c. $a\sec\theta$

d. $a\cos\theta$

30 [Intermediate] • Target: Limits involving factorable higher-degree polynomials

Numerical integration: Simpson's rule uses polynomials of degree:

a. 1

b. 3

c. 2

d. 4

31 [Intermediate] • Target: Direct substitution and algebraic techniques

FTC in higher dimensions: $\int_a^b ablaf\cdot d\mathbf{r} = f(b)-f(a)$. This is valid for:

a. any vector field

b. gradient fields (conservative) only

c. curl-free fields on simply-connected domains

d. divergence-free fields

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32 [Intermediate] • Target: Trigonometric and special limits

The substitution $u = g(x)$ in $\int_a^b f(g(x))g'(x)\,dx$ requires g to be:

a. continuous only

b. monotone only

c. C^1 with $g' \geq 0$

d. absolutely continuous

33 [Intermediate] • Target: Limits that fail to exist and the squeeze theorem

Trig substitution for $\sqrt{a^2 - x^2}$: set $x =$

a. $a\tan\theta$

b. $a\sin\theta$

c. $a\sec\theta$

d. $a\cos\theta$

34 [Intermediate] • Target: Limits with absolute value and piecewise functions

Stokes' theorem: $\iint_S (abla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$. Relates:

a. divergence to flux

b. gradient to work

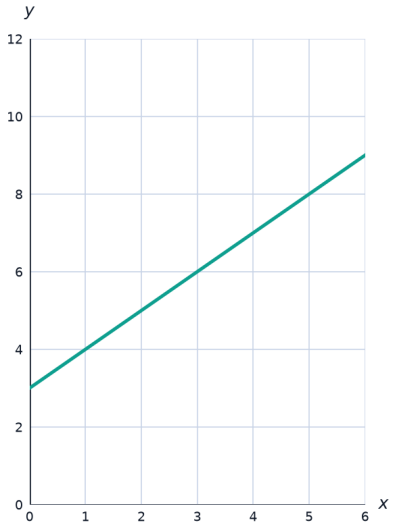
c. curl to circulation

d. Laplacian to area

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35 **[Intermediate]** • **Target: More algebraic limit techniques**

Consider the function $f(x)$ defined piecewise as: $f(x) = \begin{cases} \frac{x^2-9}{x-3}, \text{ if } xeq3; \\ k, \text{ if } x = 3 \end{cases}$ For the function $f(x)$ to be continuous at $x = 3$, the value of k must be such that the limit of $f(x)$ as x approaches 3 exists and is equal to $f(3)$. Which of the following statements is true regarding the continuity of $f(x)$ at $x = 3$ if k is set to satisfy this condition, and what is the value of that limit?



a. The function is continuous at $x = 3$ and the limit is 6.

b. The function is continuous at $x = 3$ and the limit is 9.

c. The function is not continuous at $x = 3$ regardless of the value of k.

d. The function is continuous at $x = 3$ and the limit is 3.

36 **[Intermediate]** • **Target: Visualizing a limit numerically and graphically**

Consider the function $f(x) = \frac{1}{x^2+1}$. Which of the following statements describes the behavior of the function as x approaches infinity?

a. The function approaches 0.

b. The function approaches 1.

c. The function approaches ∞.

d. The function oscillates.

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37 [Intermediate] • Target: Limits at infinity with different degree relationships

Using Stokes' Theorem, evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle -y^2, x, z^2 \rangle$ and C is the intersection of $z = y$ and the cylinder $x^2 + y^2 = 1$ (oriented counterclockwise from above).

a. $\pi/2$

b. π

c. 3π

d. $3\pi/2$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Limits requiring conjugate multiplication • Cross-Lesson: 12.01 Convergence Tests

The figure shows the graph of a function $f(x)$. What is the value of $\lim_{x \to 2} f(x)$?

a. 1

b. 2

c. 3

d. The limit does not exist

39 [Challenging] • Target: Piecewise limits requiring a parameter • Cross-Lesson: 12.02 Power Series and Taylor Series

If f is monotonic on $[a, b]$, then f is:

a. Riemann integrable

b. necessarily continuous

c. necessarily differentiable

d. not integrable if it has jumps

40 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Rolle's Theorem requires $f(a) = f(b)$ and f is:

a. differentiable on (a, b) only

b. continuous on $[a, b]$

c. both continuous on $[a, b]$ and differentiable on (a, b)

d. monotone

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41 [Challenging] • Target: Direct substitution and algebraic techniques • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Using De Moivre's theorem, evaluate $(1 + i)^9$.

a. $16 - 16i$

b. $16 + 16i$

c. $-16 - 16i$

d. $-16 + 16i$

42 [Challenging] • Target: Trigonometric and special limits • Cross-Lesson: 15.01 Double Integrals

If f is merely Riemann integrable (not necessarily continuous) on $[a, b]$, then $g(x) = \int_a^x f(t)\,dt$ is guaranteed to be:

a. continuous on $[a, b]$

b. differentiable everywhere

c. equal to an antiderivative of f

d. constant wherever f is discontinuous

43 [Challenging] • Target: Limits that fail to exist and the squeeze theorem • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Picard's theorem guarantees existence and uniqueness of IVP solution when f and $\partial f/\partial y$ are:

a. constant

b. polynomial

c. continuous on a neighborhood of (x_0, y_0)

d. analytic everywhere

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44 [Challenging] • Target: Limits with absolute value and piecewise functions • Cross-Lesson: 2.02 Continuity

Changing the value of a Riemann-integrable f at finitely many points in $[a, b]$ leaves $\int_a^b f\,dx$ unchanged because:

a. a finite set has measure zero, so it doesn't affect the integral

b. f is still continuous

c. the antiderivative is unaffected by definition

d. Riemann sums always skip finite sets

45 [Challenging] • Target: More algebraic limit techniques • Cross-Lesson: 3.01 The Derivative from First Principles

Trig substitution $\int \sqrt{4 - x^2}\,dx$ with $x = 2\sin\theta$ gives:

a. $\arcsin(x/2) + C$

b. $\frac{x\sqrt{4 - x^2}}{2} + 2\arcsin(x/2) + C$

c. $\sqrt{4 - x^2} + C$

d. $-\sqrt{4 - x^2} + C$

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